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CS-499 Computer Science Capstone 2025 C-5 (Sep - Oct)

4-2 Milestone Three: Enhancement Two: Algorithms and Data Structure

The artifact I selected for this enhancement is a travel web application built with Angular (frontend), Node.js/Express (backend), and MongoDB (database). It was originally created during CS-465: Full Stack Development. I chose to include this artifact in my ePortfolio because it highlights my ability to design, implement, and test a functional full-stack application.

This artifact demonstrates my skills in algorithms and data structures by showcasing how I can manipulate and organize application data to improve functionality. For this milestone, I enhanced the artifact by adding sorting and filtering features to the trip listing. I updated the Angular component to allow users to sort trips by price or date in ascending and descending order, as well as filter trips by destination. To support this functionality, I extended the data model with an index signature, which allowed me to dynamically access trip properties during sorting operations.

In module one, I planned to demonstrate outcomes related to designing and evaluating computing solutions using algorithmic principles. I believe I achieved these outcomes by implementing sorting algorithms, and handling data comparisons.

Through this enhancement, I learned how to extend a full stack application in a way that remains secure and reliable, and how to balance strong typing in TypeScript with the flexibility required for dynamic property access.

One challenge I faced was TypeScript strict type-checking, which initially prevented me from sorting dynamically by different trip properties, I overcome this challenge by using an index signature to the model, which allowed for more flexible data sorting. Solving this problem helped me strengthen my skills in data structures and their interaction with real world applications.